Yi Zhao

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Work Experience:

- Mar, 2024 to now — **Boya Postdoctoral Fellowship** — Advised by Prof. Dongfeng Li
  College of Environmental Sciences and Engineering
  Peking University

Education:

- Sep, 2019 to Sep, 2023 — **Ph.D.** — Advised by Prof. Zhuotong Nan
  School of Geography, Nanjing Normal University, China
  Major: Cartography and Geographical Information System

- Sep, 2016 to Jul, 2019 — **M.S.** — Advised by Prof. Zhuotong Nan
  School of Geography, Nanjing Normal University, China
  Major: Cartography and Geographical Information System

- Sep, 2012 to Jul, 2016 — **B.S.**
  School of Geography, Nanjing Normal University, China
  Major: Geographical Information System

Research Interests:

- Permafrost degradation on the Tibetan Plateau and the associated ecological, environmental, and socio-economic impacts
- Riverine landscape evolution in the permafrost regions and the aquatic water-sediment-carbon consequences
- Development of Land Surface Models for modeling the cryospheric environment under climate change

Representative Publications:

- **Zhao, Y.**, Li, D., Nitzbon, J., et al. Westerly-induced precipitation increase drives near-surface ground ice gain on the Tibetan Plateau despite warming. ***Science Advances***, 2026 (Accepted)

- **Zhao, Y.**, Nan, Z., Ji, H., et al. A hybrid modeling approach for improved simulation of thermal-hydrological dynamics in active layer on the Qinghai-Tibet Plateau. ***Water Resources Research***, 2025, 61(12): e2025WR040288.

- **Zhao, Y.**, & Li, D. (2025). Accelerated Permafrost Thaw Linked to Rising River Temperature and Widening Channels. ***Geophysical Research Letters***, 52(1), e2024GL112752

- **Zhao, Y.**, Nan, Z., Cao, Z., Ji, H., & Hu, J. (2023). Evaluation of parameterization schemes for matric potential in frozen soil in land surface models: A modeling perspective. ***Water Resources Research***, 59(6), e2023WR034644

- **Zhao, Y.**, Nan, Z., Ji, H., & Zhao, L. (2022). Convective heat transfer of spring meltwater accelerates active layer phase change in Tibet permafrost areas. ***The Cryosphere***, 16(3), 825–849

- **<u>Zhao, Y.</u>**, Nan, Z., Yu, W., & Zhang, L. (2019). Calibrating a hydrological model by stratifying frozen ground types and seasons in a cold alpine basin. ***Water***, 11(5), 985
- <u>赵奕</u>, 南卓铜, 李祥飞, 徐毅, 张凌. (2019). 分布式水文模型DHSVM在西北高寒山区流域的适用性研究. *冰川冻土*, (01), 147–157.

Fundings

- **National Natural Science Foundation of China (42501152)**, "Influence mechanism of the extreme precipitation on the hydro-thermal state of permafrost in Sanjiangyuan region", **<u>PI</u>**, 2026 to present
- **China Postdoctoral Science Foundation (2025M770375)**, "The development of thaw slumps and its impacts on sediment and soil carbon in the Tibetan Plateau", **<u>PI</u>**, 2025 to present

Services and Awards

- **"Best Oral Presentation by an Early Career Scientist"** at International Workshop on Cryosphere Changes and Their Regional & Global Impacts, Dunhuang, China, 2018.
- Service as reviewer for ***Geophysical Research Letters***, ***JGR: Earth Surface***, ***Journal of Hydrology***

Publication list:

- **<u>Zhao, Y.</u>**, Li, D., Nitzbon, J., et al. Westerly-induced precipitation increase drives near-surface ground ice gain on the Tibetan Plateau despite warming. ***Science Advances***, 2026 (Accepted)
- **<u>Zhao, Y.</u>**, Nan, Z., Ji, H., et al. A hybrid modeling approach for improved simulation of thermal-hydrological dynamics in active layer on the Qinghai-Tibet Plateau. ***Water Resources Research***, 2025, 61(12): e2025WR040288.
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- **<u>Zhao, Y.</u>**, Nan, Z., Ji, H., & Zhao, L. (2022). Convective heat transfer of spring meltwater accelerates active layer phase change in Tibet permafrost areas. ***The Cryosphere***, 16(3), 825–849
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- <u>赵奕</u>, 南卓铜, 李祥飞, 徐毅, 张凌. (2019). 分布式水文模型DHSVM在西北高寒山区流域的适用性研究. *冰川冻土*, (01), 147–157.
- Tian, S., Li, D., Regnier, P., Dean, J., Syvitski, J., Feng, M., Sha, A., **<u>Zhao, Y.</u>**, et al. Rapid changes in global river particulate organic carbon flux. (2026) ***Nature Geoscience***, 1-7.
- Tian, S., Li, D., Zhang, T., McClelland, J., Overeem, I., Lane, S., Spencer, R., Wohl, E., Sha, A., **<u>Zhao, Y.</u>**, et al. (2026) Increasing river sediment concentration and flux across the pan-Arctic. ***Nature Geoscience***, 19, 565–572.
- Zhang, G., Mu, C., Zhang, Y., Zhu, X., **<u>Zhao, Y.</u>**, et al. Quantifying the impacts of increasing light and moderate rainfall on permafrost thermal regimes over the Qinghai-Tibet Plateau: A controlled sensitivity study. (2026)

Journal of Hydrology, 667, 134926.

- Liu, G., Li, D., Zhang, T., Best, J., Elser, J., Filippelli, G., Tian, H., **Zhao, Y.**, et al. (2025) Human alterations to global riverine phosphorus fluxes to the ocean[J]. ***Science Advances***, 11(50), eady5884.
- Li, Y., Zhang, T., **Zhao, Y.**, Guo, Z., Han, P., Zhong, Q., et al. (2025). Impact of climate change and vegetation greening on sediment transport in the Yarlung Tsangpo River. ***CATENA***, 261, 109557
- Deng, R., Li, D., **Zhao, Y.**, Walling, D. E., Panagos, P., Borrelli, P., et al. (2025). Intensified rainfall overrides vegetation greening in driving erosion and carbon loss on the Tibetan Plateau. ***Science Bulletin***, 2025, 70(22), 3698–3702
- Bai, Y., Sun, S., Xu, Y., **Zhao, Y.**, Pan, Y., Xiao, Y., & Li, R. (2025). Exploring the dynamic impact of future land use changes on urban flood disasters: A case study in Zhengzhou City, China. ***Geography and Sustainability***, 6(4), 100287
- Bai, Y., Li, D., Wangchuk, S., Kettner, A., **Zhao, Y.**, Deng, R., et al. (2025). Flood complexity and rising exposure risk in High Mountain Asia under climate change. ***Science Bulletin***, 2025, 70(10), 1601-1604
- Sha, A., Li, D., Walling, D., **Zhao, Y.**, Tian, S., Chen, D., et al. (2025). Accelerated River Meander Migration on the Tibetan Plateau Caused by Permafrost Thaw. ***Geophysical Research Letters***, 52(1), e2024GL111536
- Tian, S., Sha, A., Luo, Y., Ke, Y., Spencer, R., Hu, X., Ning, M., **Zhao, Y.**, et al. (2025). A novel framework for river organic carbon retrieval through satellite data and machine learning. ***ISPRS Journal of Photogrammetry and Remote Sensing***, 221, 109–123
- Xue, Z., Wang, Y., **Zhao, Y.**, Li, D., & Borthwick, A. G. L. (2024). Impacts of spatially inconsistent permafrost degradation on streamflow in the Lena River Basin. ***Science China Technological Sciences***, 67(11), 3559–3570
- Cao, Z., Gao, H., Nan, Z., **Zhao, Y.**, & Yin, Z. (2021). A Semi-Physical Approach for Downscaling Satellite Soil Moisture Data in a Typical Cold Alpine Area, Northwest China. ***Remote Sensing***, 13(3), 509
- Ji, H., Nan, Z., Hu, J., **Zhao, Y.**, & Zhang, Y. (2022). On the spin-up strategy for spatial modeling of permafrost dynamics: A case study on the Qinghai-Tibet Plateau. ***Journal of Advances in Modeling Earth Systems***, 14(3), e2021MS002750
- Zhang, L., Ren, D., Nan, Z., Wang, W., **Zhao, Y.**, Zhao, Y., et al. (2020). Interpolated or satellite-based precipitation? Implications for hydrological modeling in a meso-scale mountainous watershed on the Qinghai-Tibet Plateau. ***Journal of Hydrology***, 583, 124629